

Arjuna JEE 2025

Maths

DPP: 3

Quadratic Equations

Q1 The quadratic equation with rational coefficient has one root $1 + \sqrt{3}$ is

- (A) $x^2 + 2x + 2 = 0$
 (B) $x^2 - 2x + 2 = 0$
 (C) $x^2 + 2x - 2 = 0$
 (D) $x^2 - 2x - 2 = 0$

Q2 The quadratic equation whose one root is $\frac{1}{2+\sqrt{5}}$ will be

- (A) $x^2 + 4x - 1 = 0$
 (B) $x^2 + 4x + 1 = 0$
 (C) $x^2 - 4x - 1 = 0$
 (D) $\sqrt{2}x^2 - 4x + 1 = 0$

Q3 If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube the other root, then a value of k is

- (A) -81
 (B) 100
 (C) 144
 (D) -300

Q4 Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

- (A) $x^2 - 18x - 16 = 0$
 (B) $x^2 - 18x + 16 = 0$
 (C) $x^2 + 18x - 16 = 0$
 (D) $x^2 + 18x + 16 = 0$

Q5 The roots of the equation $x^3 - 2x^2 - x + 2 = 0$ are

- (A) 1, 2, 3
 (B) -1, 1, 2
 (C) -1, 0, 1
 (D) -1, -2, 3

Q6

If 1, 2, 3 are the roots of the equation $x^3 + ax^2 + bx + c = 0$, then

- (A) $a = 1, b = 2, c = 3$
 (B) $a = -6, b = 11, c = -6$
 (C) $a = 6, b = 11, c = 6$
 (D) $a = 6, b = 6, c = 6$

Q7 Let r_1, r_2, r_3 be roots of equation $x^3 - 2x^2 + 4x + 5074 = 0$, then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is

- (A) 5050
 (B) -5050
 (C) -5066
 (D) -5068

Q8 Let a, b, c are roots of equation $x^3 + 8x + 1 = 0$, then the value of $\frac{bc}{(8b+1)(8c+1)} + \frac{ac}{(8a+1)(8c+1)} + \frac{ab}{(8a+1)(8b+1)}$ is equal to

- (A) 0
 (B) -8
 (C) -16
 (D) 16

Q9 If a, b, c are real and $x^3 - 3b^2x + 2c^3$ is divisible by $x - a$ and $x - b$, then

- (A) $a = -b, = -c$
 (B) $a = 2b = 2c$
 (C) $a = b, b = c, a = -2b = -2c$
 (D) None of these

Q10 If the sum of two roots of the equation $x^3 - px^2 + qx - r = 0$ is zero, then

- (A) $pq = r$
 (B) $qr = p$
 (C) $pr = q$
 (D) $pqr = 1$



- Q11** If the sum of two of the roots of $x^3 + px^2 + qx + r = 0$ is zero, then $pq =$
(A) $-r$
(B) r
(C) $2r$
(D) $-2r$
- Q12** If the sum of the two roots of the equation $4x^3 + 16x^2 - 9x - 36 = 0$ is zero, then the roots are
(A) $1, 2, -2$
(B) $-2, \frac{2}{3}, -\frac{2}{3}$
(C) $-3, \frac{3}{2}, -\frac{3}{2}$
(D) $-4, \frac{3}{2}, -\frac{3}{2}$
- Q13** Let α, β, γ are roots of the equation $x^3 + x^2(3 \sin \theta) + 4x \cos \theta - 1 = 0$, where $\theta \in R$, then maximum value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$ is:
(A) 1 (B) 3
(C) 4 (D) 5
- Q14** The two roots of an equation $x^3 - 9x^2 + 14x + 24 = 0$ are in the ratio $3 : 2$. Find the roots.



Answer Key

Q1 (D)

Q2 (A)

Q3 (D)

Q4 (B)

Q5 (B)

Q6 (B)

Q7 (B)

Q8 (C)

Q9 (C)

Q10 (A)

Q11 (B)

Q12 (D)

Q13 (D)

Q14 6, 4, -1



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